

Instance Segmentation using Mask R-CNN

Team members:

1. Suraj Thapa
2. Abhigya Malla

Description of the task:

For this project, we aim to perform instance segmentation using the Mask R-CNN architecture. Instance segmentation combines elements from object detection and semantic segmentation; it not only identifies and classifies each object within an image but also delineates its precise boundary at the pixel level. Unlike traditional object detection that provides bounding boxes, instance segmentation assigns unique masks to each detected object, even when multiple instances belong to the same class.

The task involves developing and training a model capable of identifying and segmenting specific object categories within a custom dataset. For instance, in an image containing multiple objects such as shoes, pens, and electric plugs, the model will detect each object, classify it correctly, and generate an accurate mask outlining its region.

Applications: Instance segmentation plays a crucial role in various domains, such as:

- Autonomous driving: Identifying vehicles, pedestrians, and road signs.
- Medical imaging: Detecting and segmenting anatomical structures or tumors.
- Industrial inspection: Identifying product defects or classifying manufactured components.
- Agriculture and retail: Counting and categorizing products or crops.
- Scientific analysis: Segmenting cells or satellite imagery for terrain classification.

Mask R-CNN Architecture: Mask R-CNN extends the Faster R-CNN framework by adding a parallel branch for predicting segmentation masks on each Region of Interest (RoI), alongside the existing branches for object classification and bounding-box regression. It uses a two-stage process:

1. The Region Proposal Network (RPN) generates candidate object regions.
2. The second stage classifies these regions, refines bounding boxes, and predicts a binary mask for each detected object.

This design allows Mask R-CNN to achieve both high accuracy in detection and fine-grained segmentation at the pixel level.

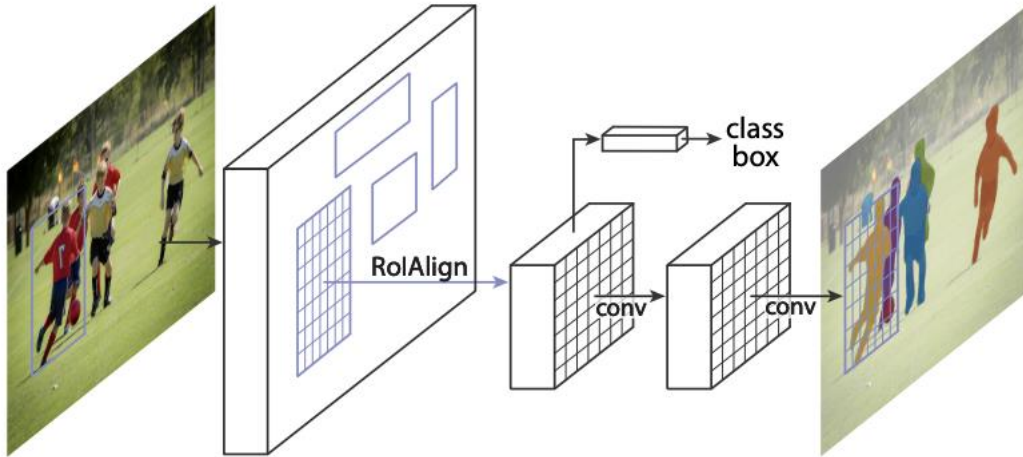


Figure 1. The Mask R-CNN framework for instance segmentation.

Description of the dataset:

We will use a custom-collected dataset consisting of at least 200 images. The dataset focuses on three object categories:

1. Shoe
2. Pen
3. Electric Plug

Data Format:

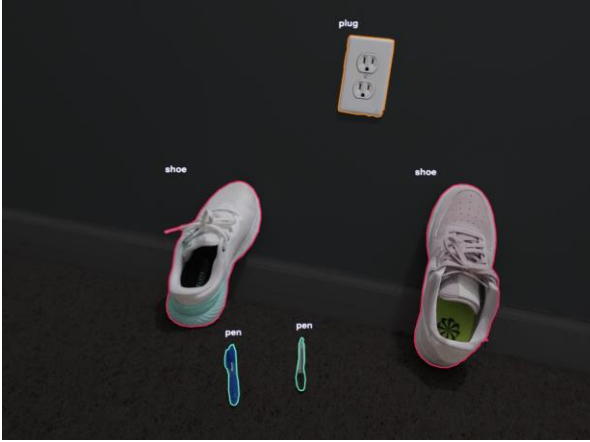
- Input: Images in .jpg or .png format.
- Annotations: JSON files containing bounding boxes, segmentation masks, and class labels for each object instance.
- Output: Segmented and labeled images where each object instance is assigned to a unique mask.

Data Source and Collection: The images will be self-captured using mobile or digital cameras under varying camera angle and background conditions to ensure data diversity.

Annotation Tool: Annotations will be created using Roboflow, a web-based annotation tool that allows users to draw polygons or bounding boxes around objects and export annotations in COCO JSON format, compatible with Mask R-CNN training frameworks.

Sample Annotated Images:

Below are example samples from the dataset annotated using Roboflow:



Img1.jpg



Img2.jpg

Figure 2. Annotated images with three labels using Roboflow.

JSON of the annotated images:

```
{
  "boxes": [
    {
      "id": "1",
      "type": "polygon",
      "label": "shoe",
      "x": "2480.5000",
      "y": "1510.5000",
      "width": "459.0000",
      "height": "997.0000",
      "points": [
        [2253,1813],[2251,1866],[2257,1897],[2272,1931],[2289,1954],[2324,1984],[2339,1993],[2388,
        2008],[2432,2009],[2480,1993],[2497,1979],[2519,1968],[2559,1930],[2578,1917],[2594,1898],
        [2604,1892],[2607,1876],[2628,1865],[2631,1859],[2645,1865],[2645,1855],[2638,1852],[2649,
        1830],[2667,1763],[2689,1719],[2698,1687],[2705,1678],[2710,1604],[2705,1573],[2707,1531],
        [2696,1504],[2672,1472],[2672,1455],[2680,1427],[2691,1351],[2693,1303],[2690,1263],[2677,
        1179],[2663,1133],[2645,1094],[2617,1056],[2596,1038],[2568,1023],[2536,1014],[2502,1012],
        [2468,1018],[2453,1025],[2428,1042],[2396,1073],[2350,1153],[2324,1235],[2317,1278],[2318,
        1354],[2329,1416],[2328,1434],[2315,1479],[2294,1591],[2284,1701]]
      ]
    }
  ]
}
```

```
},
```

```
{
```

```
  "id": "1",
```

```
  "type": "polygon",
```

```
  "label": "shoe",
```

```
  "x": "1157.0000",
```

```
  "y": "1392.5000",
```

```
  "width": "522.0000",
```

```
  "height": "791.0000",
```

```
  "points": [
```

```
[1191,1042],[1140,1092],[1090,1157],[1062,1208],[1042,1222],[1015,1232],[987,1237],[968,1247],[896,1267],[897,1276],[911,1277],[921,1271],[973,1258],[983,1258],[995,1266],[986,1294],[966,1324],[951,1369],[938,1463],[922,1515],[918,1554],[919,1601],[914,1611],[905,1664],[908,1703],[923,1739],[953,1764],[985,1780],[1036,1788],[1117,1781],[1155,1765],[1185,1739],[1220,1675],[1235,1600],[1240,1533],[1274,1438],[1302,1411],[1319,1400],[1370,1337],[1405,1268],[1418,1199],[1417,1160],[1406,1095],[1377,1038],[1348,1012],[1329,1002],[1310,997],[1287,997],[1247,1008]]
```

```
},
```

```
{
```

```
  "id": "1",
```

```
  "type": "polygon",
```

```
  "label": "plug",
```

```
  "x": "1988.5000",
```

```
  "y": "424.5000",
```

```
  "width": "301.0000",
```

```
  "height": "443.0000",
```

```
  "points": [
```

```
[1839,584],[1838,609],[1845,615],[1944,632],[1983,632],[2005,641],[2024,637],[2046,646],[20
```

```
66,643],[2076,626],[2073,614],[2084,607],[2135,283],[2139,235],[2135,229],[2110,223],[1906,203],[1898,208],[1892,219],[1886,260],[1887,278],[1880,301],[1868,377],[1869,388]
```

```
]
```

```
},
```

```
{
```

```
"id": "1",
```

```
"type": "polygon",
```

```
"label": "pen",
```

```
"x": "1260.5000",
```

```
"y": "2049.5000",
```

```
"width": "77.0000",
```

```
"height": "331.0000",
```

```
"points": [
```

```
[1222,1953],[1222,1995],[1228,2012],[1234,2016],[1243,2050],[1242,2129],[1249,2186],[1255,2207],[1270,2215],[1281,2212],[1292,2196],[1299,2154],[1294,2083],[1288,2052],[1281,1922],[1268,1884],[1255,1887],[1246,1898],[1245,1907],[1229,1915]
```

```
]
```

```
},
```

```
{
```

```
"id": "1",
```

```
"type": "polygon",
```

```
"label": "pen",
```

```
"x": "1633.0000",
```

```
"y": "1990.5000",
```

```
"width": "52.0000",
```

```
"height": "289.0000",
```

```
"points": [
```

```
[1624,1879],[1623,1961],[1607,2065],[1613,2109],[1625,2131],[1639,2135],[1648,2131],[1654,2120],[1659,2070],[1649,2012],[1654,1973],[1652,1883],[1645,1869],[1649,1859],[1646,1850],[1639,1846],[1635,1849]
```

```
]
```

```
}
```

```
],
```

```
"height": 2448,
```

```
"key": "img1.jpg",
```

```
"width": 3264
```

```
}
```

```
{
```

```
"boxes": [
```

```
{
```

```
"id": "1",
```

```
"type": "polygon",
```

```
"label": "shoe",
```

```
"x": "1463.0000",
```

```
"y": "1621.5000",
```

```
"width": "1150.0000",
```

```
"height": "613.0000",
```

```
"points": [
```

```
[891,1605],[888,1654],[892,1678],[907,1716],[916,1727],[933,1785],[950,1821],[988,1856],[1026,1874],[1081,1883],[1119,1882],[1151,1876],[1167,1884],[1155,1900],[1126,1921],[1123,1926],[1130,1928],[1207,1898],[1242,1873],[1266,1864],[1366,1854],[1541,1855],[1648,1851],[1727,1840],[1734,1835],[1759,1831],[1817,1812],[1907,1764],[1959,1719],[2011,1650],[2033,1600],[2038,1550],[2033,1515],[2009,1472],[1981,1445],[1955,1432],[1879,1419],[1714,1414],[1672,1418],[1651,1413],[1581,1412],[1575,1400.833],[1574,1362],[1553,1343],[1535,1335],[1489,1343],[1455,1373],[1440,1381],[1406,1372],[1397,1361],[1383,1358],[1372,1340],[1339,1333],
```

```
[1307,1315],[1283,1315],[1265,1323],[1188,1385],[1148,1411],[1129,1419],[1116,1433],[1062,1456],[1045,1458],[975,1485],[937,1484],[926,1490],[909,1516]
```

```
]
```

```
},
```

```
{
```

```
  "id": "1",
```

```
  "type": "polygon",
```

```
  "label": "shoe",
```

```
  "x": "1209.5000",
```

```
  "y": "652.5000",
```

```
  "width": "821.0000",
```

```
  "height": "465.0000",
```

```
  "points": [
```

```
[801,664],[799,695],[806,722],[847,790],[883,819],[920,836],[952,843],[1028,842],[1063,836],[1115,839],[1209,833],[1276,864],[1308,868],[1346,882],[1426,885],[1460,877],[1485,879],[1524,864],[1587,824],[1608,798],[1617,771],[1620,736],[1617,714],[1607,696],[1590,682],[1551,664],[1438,633],[1377,610],[1348,593],[1334,568],[1319,565],[1306,568],[1291,546],[1287,528],[1271,505],[1268,479],[1271,435],[1268,425],[1257,420],[1240,420],[1224,427],[1207,452],[1190,463],[1198,473],[1189,485],[1178,488],[1169,481],[1156,478],[1121,482],[1077,500],[1067,511],[1022,527],[997,528],[963,522],[911,493],[886,485],[869,488],[863,500],[862,570],[834,616],[820,629]
```

```
]
```

```
},
```

```
{
```

```
  "id": "1",
```

```
  "type": "polygon",
```

```
  "label": "plug",
```

```
  "x": "2355.0000",
```

```
  "y": "787.5000",
```

```
  "width": "468.0000",
```

```

    "height": "373.0000",
    "points": [
[2129,807],[2121,817],[2124,836],[2131,842],[2135,859],[2151,894],[2156,898],[2159,912],[21
70,927],[2173,951],[2180,956],[2183,970],[2189,974],[2250,951],[2321,913],[2516,807],[2547,
784],[2580,769],[2589,759],[2568,717],[2553,701],[2525,658],[2517,636],[2492,601],[2480,603
],[2454,616],[2437,634],[2425,636],[2392,655],[2390,661],[2377,664],[2373,671],[2326,692],[2
273,722],[2270,727],[2222,750],[2210,760],[2211,765],[2187,770],[2135,800]
    ]
  },
  {
    "id": "1",
    "type": "polygon",
    "label": "pen",
    "x": "501.5000",
    "y": "688.5000",
    "width": "429.0000",
    "height": "211.0000",
    "points": [
[295,583],[287,587],[287,594],[297,616],[328,635],[358,646],[364,655],[397,670],[410,672],[65
6,788],[706,794],[716,789],[710,778],[693,768],[682,756],[596,710],[531,688],[456,645],[376,6
06],[351,596]
    ]
  },
  {
    "id": "1",
    "type": "polygon",
    "label": "pen",
    "x": "612.5000",

```



```
"y": "984.0000",  
"width": "295.0000",  
"height": "162.0000",  
"points": [  
  
[469,908],[465,913],[478,936],[508,951],[529,970],[567,987],[579,984],[592,986],[605,999],[66  
8,1029],[731,1057],[746,1058],[758,1065],[760,1056],[714,1037],[681,1018],[661,1013],[656,9  
98],[639,993],[622,982],[605,960],[593,957],[589,951],[505,912],[478,903]  
  
]  
}  
],  
"height": 2448,  
"key": "img2.jpg",  
"width": 3264  
}
```

References and Attachments:

Link to paper: <https://arxiv.org/abs/1703.06870>

Link to code:

1. <https://github.com/hackassin/learnopencv/tree/master/PyTorch-Mask-RCNN>
2. <https://github.com/drewbyron/pytorch-tutorials?tab=readme-ov-file#Instance-Segmentation-with-Mask-RCNN>
3. <https://github.com/facebookresearch/Detectron>
4. <https://github.com/multimodallearning/pytorch-mask-rcnn>
5. <https://medium.com/@siromermer/pipeline-for-training-custom-mask-r-cnn-instance-segmentation-models-with-pytorch-c2b263500a18>
6. <https://christianjmills.com/posts/pytorch-train-mask-rcnn-tutorial/>
7. https://docs.pytorch.org/vision/main/models/mask_rcnn.html